

CESE323 ERC1.0

Cloud Security Envisioning & Strategy

Downloadable Student Worksheet Pack

Field	Description
Audience	Students, security engineers, cloud administrators, solution architects, and workshop participants.
Format	Printable and editable worksheet pack for instructor-led or self-paced delivery.
Primary platforms	Microsoft Defender for Cloud, Microsoft Defender XDR, Microsoft Entra, Azure, Microsoft Purview, Microsoft Foundry / Azure AI.
Primary outputs	Security goals, gap analysis, Defender for Cloud CSPM review, workload protection decisions, AI security plan, remediation backlog, operational review, and capstone strategy one-pager.
<i>This pack is designed for CESE323 Cloud Security Envisioning & Strategy workshops and can be used as a student workbook, customer discovery workbook, or evidence pack during a live environment review.</i>	

Course participation details	Student input
Student / group name	
Organisation / customer context	
Date	
Tenant / directory used	
Azure subscription / resource group scope	
Selected workload modules	☐ Defender for Storage ☐ Defender for Containers ☐ Defender for App Service ☐ Defender for AI Services

How to use these worksheets

Complete the worksheets sequentially during the workshop. Use a lab tenant or explicitly approved customer scope. Capture evidence as screenshots, exported recommendations, notes, or agreed design decisions. The final capstone worksheet consolidates the security strategy and next-step roadmap.

Worksheet	Purpose	Primary evidence
1. Strategic Security Goals and Risk Priorities	Translate business goals into measurable cloud security outcomes.	Goal statements, risk vectors, executive priorities.
2. Cloud Security Gap Analysis	Compare current-state controls against desired posture and governance expectations.	Gap list, severity, owner, target date.

Worksheet	Purpose	Primary evidence
3. Defender for Cloud CSPM Review	Use posture management concepts to review Secure Score, recommendations, compliance, and attack paths.	Secure Score notes, top recommendations, risk prioritization.
4. Workload Protection Decision Matrix	Select 2 workload protection modules and justify the implementation sequence.	Selected modules, prerequisites, scope, cost/risk notes.
5. AI Workload Threat and Guardrail Mapping	Map AI workload threats, identities, prompts, data flows, and guardrails.	AI threat map, identity controls, prompt/data controls.
6. Practical Remediation Plan	Convert findings into a controlled remediation backlog.	Backlog, change controls, rollback, validation evidence.
7. Live Environment Operational Review	Guide the live review of Defender for Cloud and Defender XDR views.	Operational findings, alerts/incidents, decisions, follow-ups.
8. Capstone Strategy One-Pager	Summarize the future-state security architecture and 30/60/90-day roadmap.	Executive one-pager and implementation roadmap.

Worksheet 1

Strategic Security Goals and Risk Priorities

Field	Details
Estimated time	35-45 minutes
Learning outcome	Define cloud security goals, business drivers, risk appetite, and measurable success criteria.
Student deliverable	A prioritized list of security goals mapped to business value, owner, metric, and urgency.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

Safety rule: Use a lab subscription or approved customer scope only. Do not upload production data, personal information, HR/legal records, secrets, keys, or confidential prompts into lab systems.

1A. Business context and security drivers

Prompt	Student response
What business outcomes must the cloud security strategy protect? Examples: customer trust, service uptime, data protection, regulatory compliance, AI adoption, geographic expansion.	
Which cloud environments are in scope? Examples: Azure only, hybrid Azure/on-premises, AWS, GCP, Microsoft 365, Microsoft Foundry, SaaS workloads.	
Which stakeholder groups must be represented? Examples: CISO, CIO, platform team, SecOps, compliance, app owners, AI governance, finance, business unit owners.	

1B. Goal-to-risk mapping

Security goal	Risk vector addressed	Business impact if unresolved	Metric / KPI	Priority
				☐ High ☐ Medium ☐ Low
				☐ High ☐ Medium ☐ Low
				☐ High ☐ Medium ☐ Low
				☐ High ☐ Medium ☐ Low
				☐ High ☐ Medium ☐ Low

1C. Risk appetite calibration

Risk domain	Current tolerance	Desired posture	Decision required
Identity privilege / admin access	☐ Low ☐ Medium ☐ High		
Data exposure and leakage	☐ Low ☐ Medium ☐ High		
Internet-facing assets and attack paths	☐ Low ☐ Medium ☐ High		
AI prompt/data risk	☐ Low ☐ Medium ☐ High		
Cloud workload	☐ Low ☐ Medium ☐ High		

Risk domain	Current tolerance	Desired posture	Decision required
misconfiguration			
Incident response readiness	☐ Low ☐ Medium ☐ High		

1D. Executive alignment statement: In 3-5 sentences, define the security strategy for this environment.

Worksheet 2

Cloud Security Gap Analysis

Field	Details
Estimated time	45-60 minutes
Learning outcome	Identify control gaps between current state, workshop objectives, and desired security posture.
Student deliverable	Gap register with severity, owner, evidence, and remediation target.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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2A. Current-state control inventory

Control area	Current state	Evidence observed	Confidence	Gap?
Governance and policy			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Identity and privileged access			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
CSPM / posture management			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Workload protection plans			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Regulatory compliance tracking			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Threat detection and incident workflow			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Data protection and DLP			☐ High ☐ Med ☐ Low	☐ Yes ☐ No
AI security controls			☐ High ☐ Med ☐ Low	☐ Yes ☐ No

2B. Gap register

Gap ID	Finding / gap description	Risk impact	Severity	Owner	Target date
G-001			☐ Critical ☐ High ☐ Med ☐ Low		
G-002			☐ Critical ☐ High ☐ Med ☐ Low		
G-003			☐ Critical ☐ High ☐ Med ☐ Low		
G-004			☐ Critical ☐ High ☐ Med ☐ Low		
G-005			☐ Critical ☐ High ☐ Med ☐ Low		

2C. Compliance and governance mapping

Requirement or standard	Relevant control / expectation	Current evidence	Control owner	Status
Internal policy / baseline				☐ Met ☐ Partial ☐ Gap
Regulatory compliance dashboard standard				☐ Met ☐ Partial ☐ Gap
Microsoft Cloud Security Benchmark				☐ Met ☐ Partial ☐ Gap
Zero Trust principle				☐ Met ☐ Partial ☐ Gap
AI governance / acceptable use				☐ Met ☐ Partial ☐ Gap

2D. Key observations: What are the top three gaps that must be addressed first, and why?

Worksheet 3

Microsoft Defender for Cloud CSPM Review

Field	Details
Estimated time	45-60 minutes
Learning outcome	Review posture management signals: Secure Score, recommendations, asset inventory, regulatory compliance, attack paths, and exposure.
Student deliverable	A prioritized CSPM observation sheet with recommended actions and business rationale.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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3A. Portal navigation and scope confirmation

Done	Checklist item	Notes / evidence
☐	Confirmed Azure tenant/directory and subscription scope.	Notes / evidence: _____
☐	Confirmed Microsoft Defender for Cloud is visible for the selected scope.	Notes / evidence: _____
☐	Confirmed Environment settings are visible to the learner or instructor.	Notes / evidence: _____
☐	Confirmed Secure Score, recommendations, inventory, workbooks, or compliance views load correctly.	Notes / evidence: _____
☐	Confirmed no live production changes will be made without explicit approval.	Notes / evidence: _____

3B. Secure Score and posture observations

Area reviewed	Observation	Risk priority	Recommended action	Evidence link / screenshot ID
Secure Score trend / baseline		☐ High ☐ Med ☐ Low		
Top recommendations		☐ High ☐ Med ☐ Low		
Attack path or choke point		☐ High ☐ Med ☐ Low		
Inventory / exposed assets		☐ High ☐ Med ☐ Low		
Regulatory compliance control gap		☐ High ☐ Med ☐ Low		
Workbook / dashboard insight		☐ High ☐ Med ☐ Low		

3C. Recommendation triage

Recommendation	Affected resource(s)	Root cause	Blast radius	Implementation complexity	Decision
			☐ Tenant ☐ Subscription ☐ RG ☐ Resource	☐ High ☐ Med ☐ Low	☐ Fix now ☐ Backlog ☐ Accept risk
			☐ Tenant ☐ Subscription ☐ RG ☐ Resource	☐ High ☐ Med ☐ Low	☐ Fix now ☐ Backlog ☐ Accept risk
			☐ Tenant ☐ Subscription ☐ RG ☐ Resource	☐ High ☐ Med ☐ Low	☐ Fix now ☐ Backlog ☐ Accept risk
			☐ Tenant ☐ Subscription ☐ RG ☐ Resource	☐ High ☐ Med ☐ Low	☐ Fix now ☐ Backlog ☐ Accept risk

3D. CSPM summary: What does the posture data tell you about the current security operating model?

Worksheet 4

Workload Protection Decision Matrix

Field	Details
Estimated time	45-60 minutes
Learning outcome	Select and justify workload protection modules based on risk, readiness, cost, and implementation feasibility.
Student deliverable	A selected 2-of-4 Defender workload plan with prerequisites, scope, and expected outcomes.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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4A. Module selection matrix

Workload module	Primary risk addressed	Prerequisites ready?	Business value	Cost / change impact	Select?
Defender for Storage	Malware uploads, suspicious access patterns, sensitive data exposure in storage layers.	☐ Yes ☐ No ☐ Partial		☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Defender for Containers	Container image risk, AKS runtime threats, Kubernetes misconfiguration, cluster exposure.	☐ Yes ☐ No ☐ Partial		☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Defender for App Service	Cloud-native application exposure, runtime threats, insecure platform settings.	☐ Yes ☐ No ☐ Partial		☐ High ☐ Med ☐ Low	☐ Yes ☐ No
Defender for AI Services	AI workload threats, prompt evidence, data leakage, unsafe model interactions.	☐ Yes ☐ No ☐ Partial		☐ High ☐ Med ☐ Low	☐ Yes ☐ No

4B. Implementation scope

Selected module	Subscription / resource group	Resources in scope	Required role / permission	Plan activation decision
			☐ Reader ☐ Contributor ☐ Security Admin ☐ Owner	☐ Enable ☐ Demo only ☐ Defer
			☐ Reader ☐ Contributor ☐ Security Admin ☐ Owner	☐ Enable ☐ Demo only ☐ Defer

4C. Cost and governance checkpoint

Done	Checklist item	Notes / evidence
☐	Budget owner approved paid plan usage where applicable.	Notes / evidence: _____
☐	Defender plan scope is limited to approved subscription/resource group where possible.	Notes / evidence: _____

Done	Checklist item	Notes / evidence
☐	Cleanup owner and cleanup date have been assigned.	Notes / evidence: _____
☐	Students understand that production or confidential data must not be used in lab uploads or prompts.	Notes / evidence: _____
☐	Any live environment plan changes are covered by customer approval and rollback plan.	Notes / evidence: _____

4D. Decision justification: Why did the group select these two workload protection modules?

Worksheet 5

AI Workload Threat and Guardrail Mapping

Field	Details
Estimated time	60 minutes
Learning outcome	Map AI workload identities, data flows, prompts, risks, and required controls for Microsoft Foundry / Azure AI scenarios.
Student deliverable	An AI security control map with identity boundaries, guardrails, monitoring requirements, and policy recommendations.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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5A. AI workload profile

AI workload element	Student response
AI service / Foundry project / model deployment name	
Primary use case	
Data sources and data classifications	
Human users / groups	
Workload identities / managed identities / service principals	
External integrations, APIs, or plugins	

5B. AI threat mapping

Threat scenario	Possible impact	Existing control	Required guardrail	Monitoring / evidence
Prompt injection or malicious prompt chaining				
Sensitive data leakage through prompts or outputs				
Over-privileged AI workload identity				
Unsafe or policy-violating model behavior				
Unapproved model, endpoint, or data connector				
Runtime alert requiring investigation				

5C. Identity and access decisions for AI

Control decision	Selected option	Justification	Owner
Authentication flow	☐ User delegation ☐ Managed identity ☐ Service principal ☐ Other		

Control decision	Selected option	Justification	Owner
Access boundary	☐ Resource group ☐ Subscription ☐ Specific resource ☐ Data-plane scope		
Privileged access model	☐ Permanent ☐ PIM/JIT ☐ Break-glass only ☐ Not required		
Conditional Access requirement	☐ MFA ☐ Device compliance ☐ Location ☐ Risk-based ☐ Workload identity control		
Secrets and key management	☐ Key Vault ☐ Managed identity only ☐ Rotation required ☐ Not applicable		

5D. AI governance statement: Define the non-negotiable security rules for this AI workload.

Worksheet 6

Practical Remediation Plan and Change Control

Field	Details
Estimated time	45-60 minutes
Learning outcome	Convert recommendations and findings into a controlled remediation plan that includes owners, approvals, rollback, and validation.
Student deliverable	A remediation backlog suitable for workshop sign-off and post-course follow-up.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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6A. Remediation backlog

ID	Finding / recommendation	Root cause	Remediation action	Owner	Due date	Status
R-001						☐ New ☐ Approved ☐ Done
R-002						☐ New ☐ Approved ☐ Done
R-003						☐ New ☐ Approved ☐ Done
R-004						☐ New ☐ Approved ☐ Done
R-005						☐ New ☐ Approved ☐ Done

6B. Change control worksheet

Change control field	Student response
Change title	
Scope of change	
Business justification	
Impacted resources and users	
Required approvals	
Implementation window	
Rollback plan	
Validation test after change	

6C. Deployment blocker analysis

Blocker / friction point	Type	Impact	Resolution path	Escalation owner
	☐ Permission ☐ Cost ☐ Policy ☐ Technical ☐ Business			
	☐ Permission ☐ Cost ☐ Policy ☐ Technical ☐			

Blocker / friction point	Type	Impact	Resolution path	Escalation owner
	Business			
	☐ Permission ☐ Cost ☐ Policy ☐ Technical ☐ Business			
	☐ Permission ☐ Cost ☐ Policy ☐ Technical ☐ Business			

6D. Remediation narrative: Explain the sequencing logic for the first three actions.

Worksheet 7

Hands-On Scenarios and Live Environment Operational Review

Field	Details
Estimated time	45-60 minutes
Learning outcome	Capture outcomes from product demonstrations, scenario simulations, alert review, and live Defender for Cloud settings walkthrough.
Student deliverable	Operational review notes, observed security signals, decisions, and follow-up action list.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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7A. Scenario demonstration capture

Scenario	Security capability demonstrated	Signal observed	Operational lesson	Evidence ID
Misconfiguration discovery				
Over-privileged identity or risky access path				
Workload protection alert or recommendation				
AI threat / prompt evidence / data risk				
Regulatory compliance or policy gap				

7B. Live environment review approval and safety controls

Done	Checklist item	Notes / evidence
☐	Customer or lab owner approved the review scope.	Notes / evidence: _____
☐	Named subscription(s) and resource groups are documented.	Notes / evidence: _____
☐	No production changes are performed without explicit approval.	Notes / evidence: _____
☐	Break-glass or instructor admin access remains available.	Notes / evidence: _____
☐	Audit trail uses named student or instructor accounts.	Notes / evidence: _____
☐	Rollback and cleanup path is documented.	Notes / evidence: _____

7C. Operational readiness review

Operational capability	Current maturity	Evidence	Improvement needed	Owner
Alert triage workflow	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			
Incident ownership and escalation	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			
Recommendation lifecycle management	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			
Compliance reporting cadence	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			
Identity governance / access review	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			
AI workload governance	☐ Initial ☐ Developing ☐ Managed ☐ Optimized			

7D. Operational review summary: What must change in the security operating model after this workshop?

Worksheet 8

Capstone Security Strategy One-Pager

Field	Details
Estimated time	60-75 minutes
Learning outcome	Synthesize the workshop findings into a concise security strategy and 30/60/90-day implementation roadmap.
Student deliverable	A one-page executive-ready strategy summary with priorities, roadmap, and success measures.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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8A. Future-state architecture principles

Principle	Design decision	Why it matters	Metric / success indicator
Zero Trust			
Least privilege identity			
Unified visibility and XDR			
CSPM and exposure reduction			
Workload-specific protection			
Secure AI adoption			

8B. 30/60/90-day roadmap

Timeframe	Priority actions	Owner(s)	Expected evidence / outcome
0-30 days			
31-60 days			
61-90 days			

8C. Capstone executive summary

Problem statement: What is the highest-priority cloud security problem?

Recommended strategy: What is the proposed security strategy?

Expected value: How will the organisation measure success?

8D. Final submission checklist

Done	Checklist item	Notes / evidence
☐	All selected worksheets are completed with student/group details.	Notes / evidence: _____
☐	Evidence IDs or screenshot references are included where required.	Notes / evidence: _____
☐	Selected workload protection modules are justified.	Notes / evidence: _____
☐	Remediation backlog includes owners, dates, and validation approach.	Notes / evidence: _____
☐	Live environment changes, if any, are approved and documented.	Notes / evidence: _____
☐	Capstone one-pager is clear enough for executive review.	Notes / evidence: _____

Worksheet 9

Optional Facilitator Marking Rubric

Field	Details
Estimated time	10-15 minutes
Learning outcome	Provide a simple scoring guide for worksheet review, formative assessment, or capstone grading.
Student deliverable	A scoring rubric that can be used by the instructor or for peer review.
Evidence to attach or reference	Screenshots, portal export, notes, risk register entries, or signed-off decisions as applicable.

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9A. Rubric

Criterion	Excellent (4)	Competent (3)	Developing (2)	Needs attention (1)
Strategic alignment	Security goals are business-linked, measurable, and prioritized.	Goals are clear and mostly linked to business needs.	Goals are present but vague or incomplete.	Goals are missing or not linked to business needs.
Technical accuracy	Uses correct Defender for Cloud, Entra, CSPM, workload, and AI security concepts.	Mostly accurate with minor gaps.	Some correct concepts but several gaps.	Major misconceptions or unsupported decisions.
Risk prioritization	Findings are clearly triaged by impact, likelihood, scope, and urgency.	Findings are prioritized but rationale could be stronger.	Prioritization is inconsistent.	No meaningful prioritization.
Operational readiness	Includes owners, change control, rollback, evidence, and monitoring.	Includes most operational details.	Operational details are incomplete.	No clear operating model.
Capstone quality	Executive-ready, concise, evidence-based, and actionable.	Clear and actionable with minor refinements needed.	Partially actionable but lacks clarity.	Not suitable for stakeholder review.

9B. Score summary

Student / group	Total score / 20	Strengths	Priority improvement

Facilitator comments
